

Shock detection and capturing methods for high order Discontinuous-Galerkin Finite Element Methods

Abhishek Sheshadri*, Antony Jameson†

Department of Aeronautics and Astronautics, Stanford University, Stanford, CA, 94305

32nd AIAA Applied Aerodynamics Conference

High-order Discontinuous Galerkin (DG) methods have shown a lot of promise in being able to provide high accuracy and efficiency as well as the flexibility to handle complex unstructured grids. However in order to solve compressible flow problems effectively, we need to be able to detect discontinuities/shocks in the flow and capture them effectively such that the solution accuracy is not affected elsewhere. Traditionally finite difference and finite volume schemes have used a variety of physical sensors. Such physical sensors are not best suited to distinguish between different flow phenomena in more complex viscous flows involving boundary layers, vortices etc. Also, owing to the availability of higher resolution in each element/cell in these high order finite element methods, it would be optimal to detect shocks/discontinuities through a mathematical procedure inside each element and capture the shock inside a single element¹²³⁴. While there has been considerable progress in this direction, robust shock detection across the broad spectrum of compressible flow problems still remains an issue. While we have been able to get some of these methods to work quite well for certain cases, they need to be fine-tuned to each problem and there is lack of rigor in how one should do the same. Klöckner also points out certain problems with methods which infer the presence of discontinuities using just the decay (or not) of the highest (few) modal frequencies. The most important issue is that the modal co-efficient decay is not uniform for a small finite resolution and the situation is only worse in polynomial bases compared to Fourier spectra. All this amounts to unreliable performance in complex flows, with inadequate separation of scales between shocks and other flow features like vortices and boundary layers. These issues make it hard to build a generic solver for inviscid and viscous compressible flow computations.

In this paper we explore and present a new approach in this direction, wherein we use the ideas from edge-detection methods⁵⁶⁷ used frequently in the fields of signal processing and imaging. These methods are used to detect edges in images/signals directly from the Fourier spectral data. This can be extended to the case of polynomial modal data as well. We propose the use of the method of concentration, which when combined with non-linear enhancement provides a rigorous and simple way to distinguish between a point of discontinuity and one which is not. Here we present some ideas of how this method can be effectively and efficiently adapted to a high-order CFD setting. Along with shock detection, we also consider and discuss a few shock-capturing strategies. While addition of artificial viscosity seems to be the *modus operandi* in this regard, in our experience it is harder to capture high Mach number shocks or multiple shocks simultaneously, especially with higher order polynomial interpolation. So we explore a more direct idea: one of order reduction, but in a fashion rather amenable to these finite-element methods, wherein we reduce the order of polynomials used in the element temporarily using a modal truncation. This idea is very easy to implement, merges very easily with the shock detection mechanism and does not require any changes to mesh storage structures. We show how to adapt this combination successfully in the context of the particular class of high-order DG methods which we use called Flux Reconstruction schemes⁸⁹ implemented in our code package we

*Ph.D. Candidate, Department of Aeronautics and Astronautics, Stanford University; abisheks@stanford.edu; (650) 561-5081

†Thomas V. Jones Professor of Engineering, Department of Aeronautics and Astronautics, Stanford University, AIAA Member

call HiFiLES.¹⁰ We also discuss computational costs and how the method can be efficiently implemented on a multi-CPU-GPU platform.

I. Introduction

Capturing discontinuities in numerical solutions of PDEs has been a subject of research ever since the development of such methods itself. While there have been many established methods in the context of finite difference and finite volume schemes, shock detection and capturing pose a new challenge in the case of high-order DG methods. These methods are designed to be suitable for computations on complex unstructured grids without changes in formulation. So it is important to preserve this flexibility while designing any tools like shock capturing, mesh adaptation etc. While most of the methods extended from the approaches used in finite difference and finite volume methods work well in 1-D, they often suffer in 2-D and 3-D on unstructured grids and there have been numerous such efforts which have been validated for 1-D but not extended to multidimensional cases successfully. In finite volume methods, the techniques developed for 1D can be easily extended to multiple dimensions due to the fact that we only need to solve multiple 1-D Riemann problems in 2-D and 3-D and the one-dimensional nature is preserved due to the use of just cell averages. In the case of high-order finite element methods however, the local structure is complex and it is not possible to split it into uni-dimensional Riemann problems. As a result, recent approaches have focused more on developing techniques which can be directly or easily extended to multiple dimensions and various element shapes.

Persson and Perraire¹² introduced the idea of sub-cell shock capturing which exploits the higher resolution and information content available inside an element/cell in these methods (compared to finite volume methods) to capture the shock in a single cell instead of resolving it across multiple elements like in FVMs. They also proposed an effective mathematical approach to detect shocks using the polynomial modal co-efficients of the projected solution. While we have been able to use this method with reasonable success, especially in triangular meshes, as Klöckner et al.³ show, the mathematical theory is not rigorous and the parameters of the method have to be chosen through experimentation for each problem which makes it hard to adapt this method to a generic solver. Klöckner et al.³ suggest certain improvements/alternatives to this method which we feel are computationally intensive and the extensions to multiple dimensions is ambiguous.

The focus of this paper is to explore the idea of employing the concentration method⁵⁶⁷ used in edge detection in signal processing for shock detection. It is a method used to directly determine the presence of edges and also obtain the jump values at these edges/discontinuities from the Fourier data. The polynomial modal co-efficients are similar to the Fourier/spectral data used in these methods and this method can be extended fairly easily to polynomials bases. The idea behind this method is to use the decay of Fourier co-efficients as an indication of shock; but instead of just looking at the highest order modes (like in Persson's method), the concentration method uses a more rigorous approach to distinguish between a point of discontinuity and one that is not. This method has also been successfully adapted in the field of imaging wherein it is used in 2D and 3D contexts successfully.

Once the shock is detected in an element, there are several methods one can use to capture it effectively without compromising on the accuracy away from the shock and also to reduce or eliminate the resulting oscillations due to the Gibbs phenomena. The most commonly used method in this field is the use of artificial viscosity to resolve the shock. The general trend is to introduce artificial viscosity in a straightforward way in addition to the physical viscosity and have the artificial viscosity co-efficient(s) vary smoothly across the domain. In addition, it has been observed that discontinuities in this artificial viscosity co-efficient itself might introduce oscillations of the gradients⁴ and we need to take extra care to enforce continuity of this co-efficient to be able to reduce oscillations. However we have observed that enforcing continuity of this co-efficient alone does not completely eliminate the oscillations and additional measures like flux limiting might be necessary.

This approach of using artificial viscosity works well at low orders and low Mach numbers. In certain cases with strong shocks and at high orders, we found it hard to be able to add the right amount of dissipation to prevent the runs from becoming unstable or reaching states with negative pressure or density. So we also re-investigate order reduction as a possible option for such cases.

The method we use for flow computations is called Flux Reconstruction (FR) and is a variant of the DG-FEM methods⁸⁹. In this method, we can recover DG and Spectral Difference methods as special cases by varying a single parameter. Variation of this parameter also allows us to increase stable time step limits. The method is also very flexible in the fluxes that can be used and is highly parallelizable. We use an explicit time-stepping that allows us to port the code onto a multi-CPU-GPU platform which in order to be able to run tedious viscous problems. Extensive studies about the stability in linear problems, dispersion and dissipation have been performed by earlier authors. Though the results presented in this paper are in the context of FR methods, the shock detection and capturing ideas we use should be adaptable to most DG-based high-order methods.

II. Shock detection: Concentration method

A. Concentration method in 1D

The concentration method, proposed by Gelb, Cates and Tadmor⁵⁶⁷ is a general framework for recovering edges in piecewise smooth functions with finitely many jump discontinuities. The approach is based on two main aspects: localization using appropriate concentration kernels and separation of scales by nonlinear enhancement.

Let $[f](x) = f(x^+) - f(x^-)$ denote the jump at a point x . For a 2π periodic function f with a single jump discontinuity, the Fourier co-efficients are given by

$$\hat{f}_k = [f](c) \frac{e^{-ikc}}{2\pi ik} + \mathcal{O}\left(\frac{1}{k^2}\right)$$

Let $S_N(f)$ be the spectral projection of f (over N modes) i.e.

$$S_N(f) = \sum_{k=-N}^N \hat{f}_k e^{ikx}$$

Then using the concentration property of a Dirichlet Kernel localized at c , we can write

$$\begin{aligned} \frac{\pi}{N} S_N(f)' &= \pi \sum_{k=-N}^N \frac{ik}{N} \left\{ [f](c) \frac{e^{ik(x-c)}}{2\pi ik} + \mathcal{O}\left(\frac{1}{k^2}\right) \right\} \\ &= \frac{[f](c)}{2N} \sum_{k=-N}^N \left\{ 1 + \mathcal{O}\left(\frac{1}{k}\right) \right\} e^{ik(x-c)} \\ &= [f](x) + \mathcal{O}\left(\frac{\log N}{N}\right) \end{aligned} \tag{1}$$

As we can see, if N were large, we can expect this to recover the jumps or edges at a point x . However this convergence is just $\mathcal{O}\left(\frac{\log N}{N}\right)$ which is very slow and for the small ' N ' or resolution used in the CFD methods, we cannot expect this term to be negligible. The concentration method is basically a tool to concentrate near a point of discontinuity so as to be able to speed up this convergence. Let K be a concentration kernel such that

$$K_N^\sigma * f \equiv K_N^\sigma * S_N(f) = i\pi \sum_{k=-N}^N \text{sgn}(k) \sigma\left(\frac{|k|}{N}\right) \hat{f}_k e^{ikx}$$

where $\sigma(\cdot)$ are concentration factors. This concentration kernel is just a conjugate Fourier partial sum (of f') with additional intelligently chosen concentration factors which depend on the modal frequency. A variety of

concentration factors are available for this purpose and are discussed in detail in the references mentioned.

A simple non-linear enhancement can be used to further amplify the separation of scales between the edges/discontinuities and the smooth regions. For example, consider

$$N^{p/2}(K_N \star f(x))^p \sim \begin{cases} N^{-p/2}, & \text{at a smooth point } x \neq c \\ ([f](c))^p N^{p/2}, & \text{at a discontinuity } x = c \end{cases} \quad (2)$$

By choosing a $p > 1$ (we typically choose 2 or 3), we can therefore enhance the separation between the vanishing scale at the smooth points and growing scale at the jumps. This can be formalized as choosing the enhanced kernel

$$K_{N,J}^p \star f(x) = \begin{cases} K_N \star f(x), & \text{if } N^{p/2}|K_N \star f(x)|^p > J. \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

where p is the enhancement exponent and J is an appropriately chosen threshold. This additional step becomes very important for smaller values of N if we want a good separation of scales between shocks and vortices, for example.

Now, although we presented this approach using Fourier modes, this idea can be extended to polynomial modes (especially those derived from Chebyshev and Jacobi polynomials).⁶ The method and the results are very similar. We hereby summarize how to use the method of concentration as a shock detection mechanism in a high order finite element formulation in 1D. We assume the method uses a polynomial basis derived from Jacobi or Chebyshev polynomials in each element. We later discuss how to extend this idea to higher dimensions.

1. As a pre-processing step, at each solution or nodal point x in the element, compute $\frac{\pi}{N} \sqrt{1-x^2} \sigma(\frac{|k|}{N}) P'_k(x)$ for $k = 1, 2, \dots, N$ where N is the number of solution points (or modes) in the element. Since all our elemental computations are performed in a reference or a parent domain, these solution points are fixed and so this would just be a single $N \times N$ matrix. Let us call this the concentration matrix.
2. In each element, convert the element-local nodal solution of a particular representative quantity (say density) to modal form using the Vandermonde matrix.
3. Use the modal co-efficients to compute, at each solution point x ,

$$K_N^\sigma * S_N(f) = \frac{\pi}{N} \sqrt{1-x^2} \sum_{k=1}^N \sigma(\frac{|k|}{N}) \hat{f}_k P'_k(x)$$

Note that this would be a simple matrix-vector multiplication of the concentration matrix and the modal co-efficients. This is a $\mathcal{O}(N^2)$ computation in each element. Since most of the elemental operations in this method are of the same complexity (including computing modal co-efficients from nodal solution), we do not increase the computational complexity of the method.

4. Evaluate the enhanced kernel and identify those points for which it is non-zero as points of discontinuity.
5. If one or more points in an element have been identified as being points of discontinuity, we can then filter the solution in this element using any suitable method - addition of artificial viscosity, polynomial order reduction (corresponding to a sharp modal filter), exponential filter etc. As an alternative, we can interpret the maximum value of the enhanced kernel in an element as a “shock sensor” value similar to the method proposed by Persson and Peraire.

B. Extension to 2D and 3D

The concentration method is used successfully in 2D and 3D as well in fields of edge detection in images (for MRI etc). In 2D, in case of Fourier data, the method is extended to the square domains by considering orthogonal 1D slices and computing slice-based Fourier co-efficients as follows:

$$\tilde{f}_k(y_\lambda) = \sum_{l=-N}^N \hat{f}_{k,l} e^{il y_\lambda} \quad \text{and} \quad \tilde{f}_l(x_\nu) = \sum_{k=-N}^N \hat{f}_{k,l} e^{ik x_\nu}$$

where $y_\lambda = -\pi + \frac{\pi\lambda}{N}$ and $x_\nu = -\pi + \frac{\pi\nu}{N}$ where $\lambda, \nu = 0, 1, 2, \dots, 2N$

Using these slice based Fourier co-efficients, 2 edge maps are created: $S_N^\sigma[f](x, y_\lambda)$ and $S_N^\sigma[f](x_\nu, y)$. These edge maps can then be used to determine points of discontinuities. In order to determine the orientation of the shock, additional steps have to be taken but we do not consider them here as it is not required here. However note that this method also allows us to use solution reconstruction methods inside the elements if need be.

While this idea can be directly adapted in tensor product elements like quadrilaterals and hexagons in our finite-element formulation, we prefer to use an even simpler approach summarized below which has worked well for all our cases.

In a tensor product element like quads, each row and column of solution/nodal points can be considered as a 1D slice and a 1D modal portrait can be computed using the 1D-concentration method for each such slice. The maximum value of the enhanced kernel among all points in all slices can then be assigned as the value of the elemental ‘‘shock sensor’’ this to the element. The element can then be marked for special treatment, whether it is adding dissipation or reducing order explicitly or any other method of choice. The projection of a discontinuity oriented in a direction other than along x,y or z axes would still have a discontinuity along one or more of these directions which enables us to use this simple approach.

This simple strategy trades the accuracy of location and orientation of the shock for less computational time. Since our shock capturing methods are based at an element level, we do not profit from these calculations. Unlike the traditional applications of concentration method where it needs to be applied only once to determine the edges in an image, we need to perform this operation every few time steps and we feel that this compromise is very profitable. On triangles and tetrahedra, this particular approach of considering rows and columns (or x and y slices) is not directly applicable as the number of points in a row decreases in a triangular fashion. However we believe we can consider the nodal points along the three edges as 1D slices and take the same approach. In this paper we present results only for quads.

III. Shock Capturing Methods

Persson and Peraire in their articles on shock capturing¹² discuss a variety of shock capturing methods including their ease of applicability to high order finite element methods. They conclude that artificial viscosity addition is the most suitable method in the context of these high order methods. We have used the method this method with reasonable success in our codes. From our experience, we found a few drawbacks with this method. We found that the choice of the artificial viscosity co-efficients is not entirely straightforward and requires quite a bit of trial and error in order to add enough dissipation to capture the shock but not superfluous amounts that would damage the accuracy of the solution. Capturing high Mach number shocks has been particularly hard since the density and pressure can easily go negative in such cases if the right amount of dissipation is not added. Another issue is that enforcement of continuity of the artificial viscosity co-efficients across element boundaries (as suggested by Barter and Darmofal⁴ to reduce oscillations of the solution near element boundaries) increases the footprint of the added dissipation. Lastly, while this method is very easy to append to a viscous code, it requires us to run viscous routines even for inviscid compressible flow simulations. This adds a fair bit of computational time, can reduce the maximum stable

time step and it adds the complexity of handling viscous boundary conditions for an inviscid problem.

In view of these issues, we use modal order reduction as an alternative for artificial viscosity based methods especially for inviscid simulations, high Mach number cases and certain other difficult cases with multiple shocks of different strengths. Although, this is the same as reducing the order of the interpolating polynomial in the element, we do not change the number of flux points or solution points inside the element avoiding any changes in mesh or code structure for these elements. In our implementation, whenever necessary, we truncate the higher order modes inside the element and convert the truncated modal solution back to nodal form. However, the reconstruction process that follows remains to be of the same order. This avoids having to change the correction polynomials, number of flux/solution points and forming p-mortar elements (to enforce continuity between two different order elements). However this comes at the expense of having to perform order reduction in the element repeatedly at some frequency. As we use an explicit method which is amenable to high parallelization across a multi-CPU-GPU platform, this elemental operation is parallelized which alleviates the issue to a good extent. Moreover, this setup can be used as a platform for any other modal filters which we are currently experimenting with. The major advantage of order reduction is that it is a more direct process and can ensure non-negativity of density and pressure which is in general one of the major obstacles in compressible flow simulations, especially with high order methods.

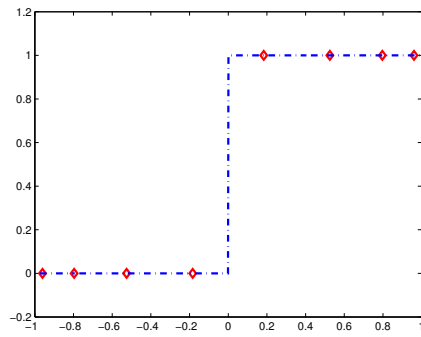
IV. Results

We tested the concentration method on all the 1D test cases Klockner³ uses in his paper. The value of the enhanced kernel is highest for the step discontinuity and is much lower for C0-kink, C1-kink and other continuous cases. Also the concentration kernel was not affected by having even or odd number of modes. Figure 1 shows the performance of the concentration method for some of the cases considered by Klöckner et al. We have used a seventh order element but results were very similar for a sixth or eight order element. We have not yet clipped the kernel at a J_{crit} . We can see from the figure that the enhanced kernel is significantly higher for the discontinuous cases (even when the step is close to the element boundary) than for the continuous cases. We tested it for several other cases not shown in the figures including the case of a constant plus a white noise. The kernel values are much smaller than those for the discontinuous cases. This shows that this method is capable of reliably separating the discontinuous cases from the rest.

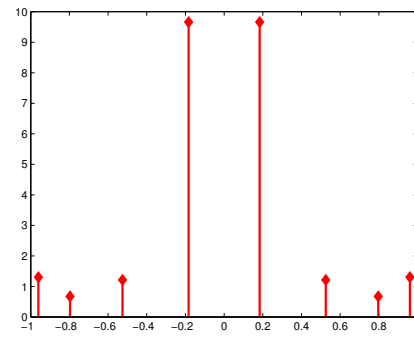
In general the value of the enhanced kernel depends on the polynomial order of the method, strength or magnitude of the jump and the location or orientation of the jump inside the element. It increases with increasing order and jump strength if all else remains the same. However, in almost all inviscid and viscous cases we simulated, there is a clear distinction between the regions with shocks and the continuous regions. As an alternative to specifying a J_{crit} as a sharp cutoff to distinguish between elements with and without shock, one could apply varying artificial viscosity over a range of the enhanced kernel values just like in other methods. Since we use 1D slices to compute the concentration kernels in elements, once the polynomial order of the scheme is decided, a simple 1D experiment can be performed to get a reasonable idea of the required range over which dissipation needs to be added. Also, in case a simulation goes unstable or if the density or pressure become negative, one can just plot the “shock sensor” to clearly see what ranges of the sensor need to be considered for adding additional dissipation. Now we move on to present some 2D results for inviscid and viscous cases. All results with the concentration method are on quadrilateral meshes. We are in the process of extending the concentration method to triangles. As of now we use Persson and Peraire’s method for triangular elements.

Meshes

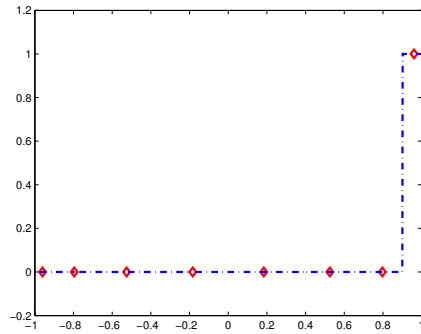
Figure 2 shows the meshes being used for the inviscid and viscous cases respectively. We are using the same meshes that we use for many of our other computations in order to observe the performance of the method on them, instead of creating a mesh tailored for these cases. We could have captured finer shocks using finer meshes, but using very fine meshes goes against the motive of using high order methods, especially in inviscid cases. So we use high order ($p = 3$ to $p = 6$) elements and relatively coarse meshes. However, since the order of the solution is reduced near the shock (either explicitly or by addition of artificial dissipation), the shock structure is coarse in the larger elements away from the body.



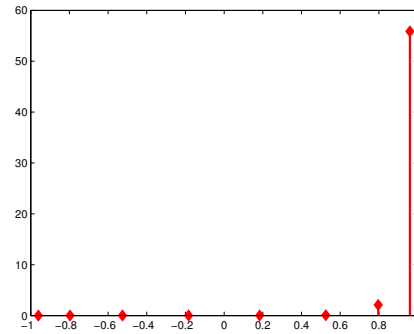
(a) Step at $x = 0$



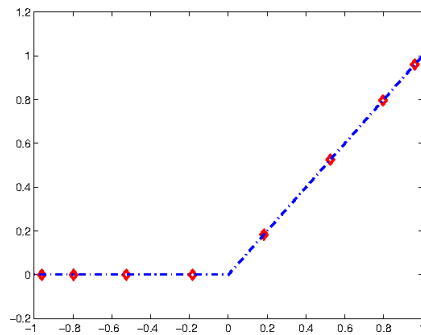
(b) Enhanced kernel portrait for (a)



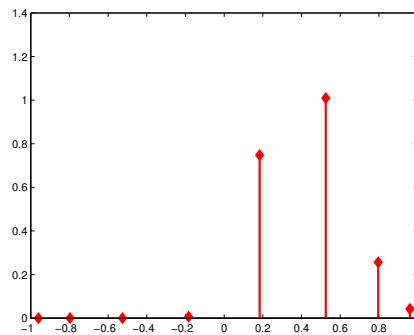
(c) Step at $x = 0.9$



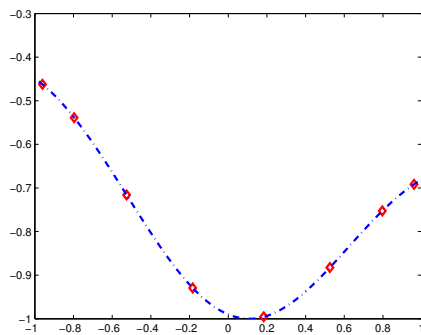
(d) Enhanced kernel portrait for (c)



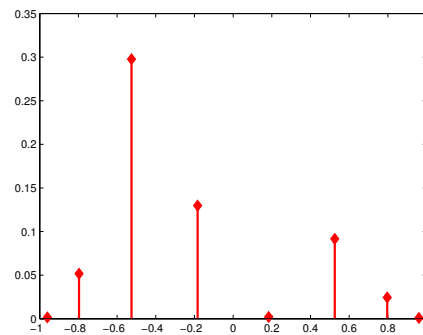
(e) C0-kink at $x = 0$



(f) Enhanced kernel portrait for (e)



(g) $\cos(3 + 1.3\sin(x))$



(h) Enhanced kernel portrait for (g)

Figure 1: Figure shows the portraits of the enhanced kernel (before clipping it below a threshold) for various cases

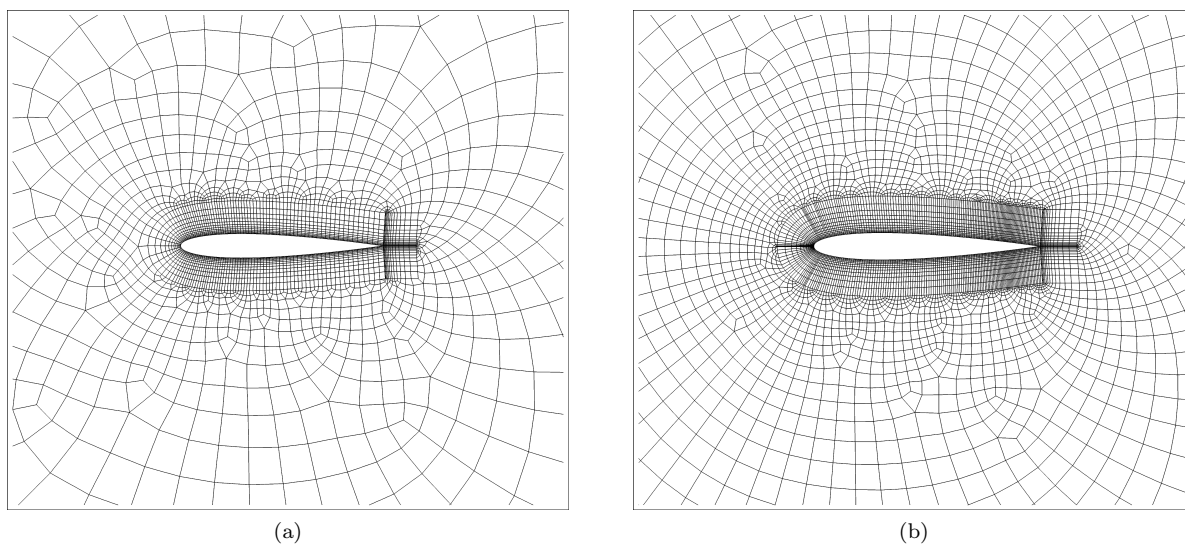


Figure 2: (a) and (b) show the meshes used for inviscid and viscous cases. (a) has around 4000 elements and (b) has close to 9000 elements

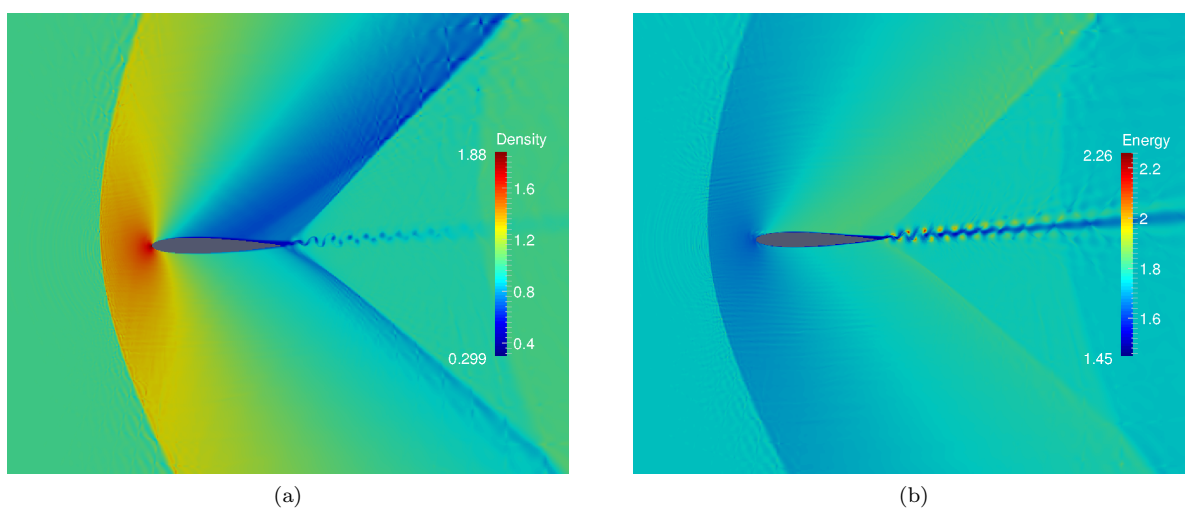


Figure 3: (a) and (b) show density and energy contours respectively for viscous flow at $M = 1.2$ over a NACA 0012 airfoil at 5° AoA with polynomial order 6 (49 points in each element)

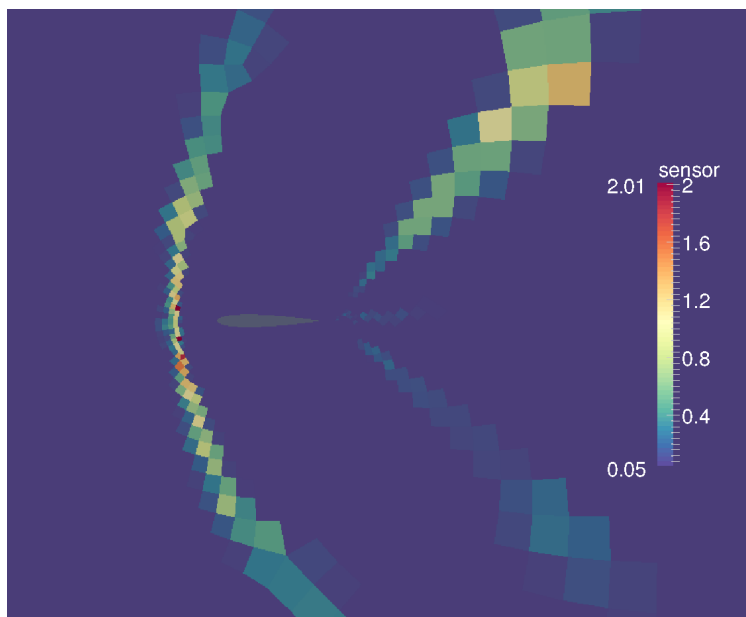


Figure 4: Figure shows the elemental shock “sensor” for the $M = 1.2$ viscous case shown in figure 3. The shock sensor is just the maximum value of the enhanced kernel in each element

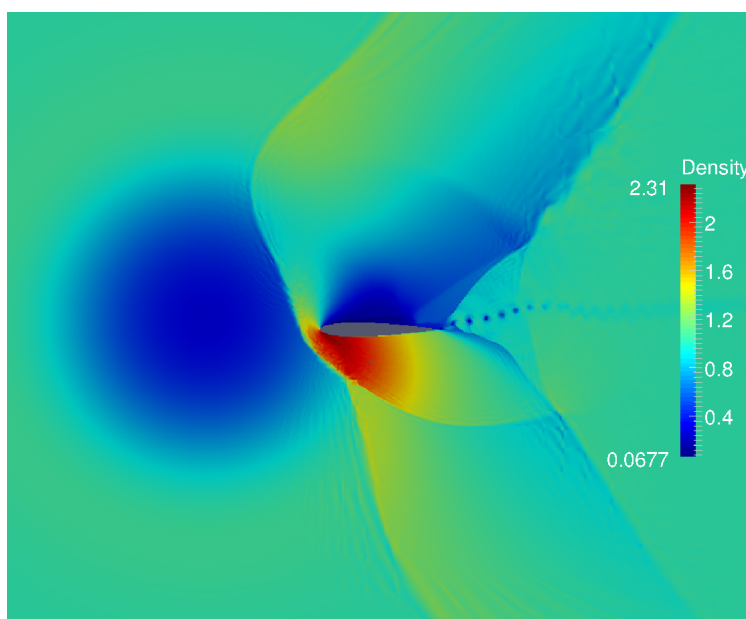


Figure 5: Figure shows the interaction between the shock system over the Mach 1.2 viscous case discussed above and a strong isentropic vortex

A. Viscous flow

We present the performance of this shock detection and capture technique using an inviscid as well as a viscous case. Figure 3 shows the flow around a NACA 0012 airfoil at a Mach number of 1.2 at 5° angle of attack. The flow is at Reynolds number of 60000 and we have used 6th order polynomial interpolation in the elements for the computation. There is a bow shock in front of the airfoil and we see fish-tail shocks at the trailing edge. We can also see boundary layer formation and a Λ -shock structure on the upper side of the airfoil. Here we have used simple modal order reduction in elements with shock sensor value above a threshold.

Figure 4 shows the elemental shock sensor values. We can see the shock sensor is able to distinguish between shocks and other smooth regions enabling the structure of the vortices and boundary layer to be preserved. This shows an advantage over other methods which suffer in making this distinction, especially those with physical shock indicators.

We also present an additional viscous case which is meant only to show the ability of the methods discussed in the paper to work well in a complex scenario and not as any verification or validation case. We superimpose a strong isentropic vortex of the form proposed by Inoue and Hattori¹¹ over the steady state flow field of the Mach 1.2 case discussed above. When this vortex reaches the airfoil, it bends the bow shock in addition to creating more weak shocks. It also bends the vortex trail. This is shown in figure 5.

B. Inviscid flow

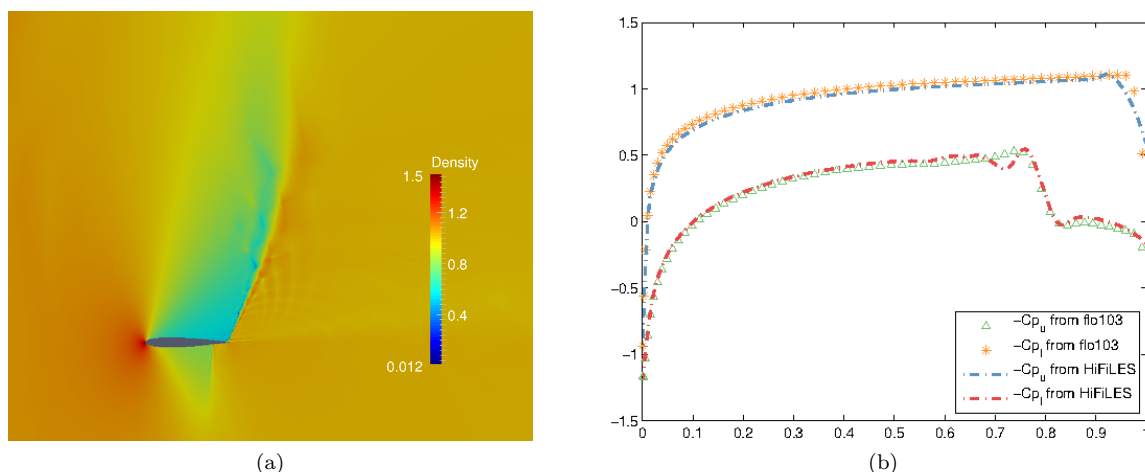


Figure 6: (a) shows density contours for inviscid flow at $M = 0.9$ over a NACA 0012 airfoil at 5° AoA run with polynomial order 3 (16 points per element). The mesh has approximately 4000 elements (b) shows a comparison of C_p plots obtained for this case from our code and those from Flo103 code written by Antony Jameson

Figure 6 shows density map for a Mach 0.9 inviscid flow over a NACA 0012 airfoil at 5° angle of attack using third order polynomials in the elements. We have used order reduction for capturing the shock. Punctuated oscillations can be seen around the shocks. We believe that the coarser mesh and the lack of inherent viscosity make these oscillations more severe than the viscous case. Part (b) of the figure 6 shows a comparison of the pressure co-efficient obtained from this computation with that from the Flo 103 code written by Antony Jameson. (We had to smooth the C_p plot to eliminate some small scale oscillations around the shock). The C_p distributions agree with each other reasonably to most extent except in front of the shock on the lower side of airfoil. We were able to reproduce the results using artificial viscosity addition for some of the cases we experimented with. However, we had to supplement it with order reduction beyond a large value of the shock sensor where the highest amount of artificial dissipation is added.

V. Summary

The idea of using the concentration method used for edge detection was explored and suggestions for effectively employing it as a shock detection tool in high order finite-element type methods were presented. Alternative efficient methods of shock capturing were also discussed. It was shown that the concentration method is promising in its ability to identify discontinuities, even in the presence of multiple shocks, vortical features and boundary layers, thus enabling us to simulate unsteady compressible viscous flows accurately. This is very encouraging from the perspective of using high order methods for flow calculations as well. An efficient black-box algorithm for the use of concentration method in quadrilateral or tensor product elements was also presented.

The extension of this method to triangles and to 3D elements is currently being worked on and will be presented in future articles. We are also working on developing more reliable filtering or shock capturing techniques which do not directly reduce the order but are also parameter-free. Finally, it is worthwhile mentioning that the cases we presented in this paper were all started with a uniform flow directly imposed on a stationary airfoil. In such cases, the shocks initially form at locations different from their final steady state locations and move to the final positions. All this is captured well by the shock capturing system we have discussed so far. This shows a promise of being able to capture moving shocks and shocks on moving meshes which is a part of our current studies.

Acknowledgments

The authors would like to acknowledge the support for this work provided by the Stanford Graduate Fellowship program, National Science Foundation (grant number 1114816) and Air Force Office of Scientific Research (grant number FA9550-10-1-0418).

References

- ¹Persson, P. and Peraire, J., "Sub-Cell Shock Capturing for Discontinuous Galerkin Methods," Tech. rep., 2006.
- ²Persson, P., "Shock Capturing for High-Order Discontinuous Galerkin Simulation of Transient Flow Problems," 21st aiaa computational fluid dynamics conference proceedings 2013-3061, 2013.
- ³A. Klöckner, T. W. b. and Hesthaven, J. S., "Viscous Shock Capturing in a Time-Explicit Discontinuous Galerkin Method," Tech. rep., 2010.
- ⁴Barter, G. and Darmofal, D., "Shock capturing with PDE-based artificial viscosity for DGFEM: Part I. Formulation," *Journal of Computational Physics*, 2010.
- ⁵Gelb, A., "Detection of Edges in Spectral Data," *Applied and Computational Harmonic Analysis* 7, 1999.
- ⁶Gelb, A. and Tadmor, E., "Detection of Edges in Spectral Data II: Non Linear Enhancement," *SIAM J. NUMER. ANAL.*, 2000.
- ⁷Gelb, A. and Cates, D., "Detection of Edges in Spectral Data III: Refinement of the Concentration Method," *J. Sci. Comput.*, 2007.
- ⁸Castonguay, P. V. P. and Jameson, A., "A New Class of High-Order Energy Stable Flux Reconstruction Schemes," *J. Sci. Comput.*, 2010.
- ⁹D. M. Williams, P. Castonguay, P. E. V. and Jameson, A., "Energy Stable Flux Reconstruction Schemes for Advection-Diffusion Problems on Triangles," *Journal of Computational Physics*, 2013.
- ¹⁰Manuel R. López-Morales, Abhishek Sheshadri, K. A. J. C. T. E. D. M. J. R. J. W. D. W. D. F. P. P. A. J., "Verification and Validation of HiFiLES: a High-Order LES unstructured solver on multi-GPU platforms," 2014.
- ¹¹Inoue, O. and Hattori, Y., "Sound generation by shock vortex interactions," *Journal of Fluid Mechanics*, 1999.